

Contraction Theory Based Control Under Input Saturation–LMI Approach [★]

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Abstract: Besides the traditional Lyapunov stability theory, contraction theory offers a interesting perspective on the stability of nonlinear systems. It allows us analyze the system stability by examining the convergence of trajectories. The recent development of the contraction theory based control has introduced computational efficient methods for obtaining contraction metrics using linear matrix inequalities (LMIs). In this paper, the contraction theory based control strategy is extended to account for input saturation by incorporating saturation constraints into the LMI framework for contraction metric calculation. The proposed method calculates the contraction metrics while ensuring the satisfaction of input saturation constraints and uses as a feedback mechanism to adjust the control inputs. The proposed method is demonstrated its effectiveness through numerical simulations via **Lorenz system**.

Keywords: Contraction theory, input saturation, linear matrix inequalities (LMIs)

NOTATION

In this paper, we use the following notation:

- $:=$ denotes *defined as*.
- $(\cdot^\top)$ denotes the transpose of a matrix or a vector.
- $\mathbf{x} := [x_i]_{i \in \{1, \dots, n\}} \in \mathbb{R}^n$ denotes the state vector.
- $\mathbf{A} := [a_{ij}]_{i, j \in \{1, \dots, n\}} \in \mathbb{R}^{n \times n}$ denotes a matrix.
- $\lambda_i(\mathbf{A})$, $i \in \{\max, \min\}$ denotes the maximum and minimum singular value of $\mathbf{A}$, respectively.
- $\mathbf{I}_n$ denotes the identity matrix of size n and $\mathbf{0}_{n \times m}$ denotes the zero matrix of size $n \times m$.
- sym denotes the symmetric part of a matrix, i.e., $\text{sym}(\mathbf{A}) := \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ (see Tsukamoto et al. (2021b)).

1. INTRODUCTION

1.1 Motivation

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- Contraction theory introduction
- Limitation 1 (computational cost)
- Recent development (LMI)
- Limitation 2 (input saturation)
 - use penalty term to prevent excessive control input
- saturation handling in LMI

To the best of our knowledge, there has been no previous work that considers the input saturation in LMI frameworks for contraction metrics. Hence, the main objective of this paper is to extend the contraction theory based control strategy to account for input saturation simultaneously in the LMI framework.

1.2 Contributions

The main contributions of this paper are

- The contraction metric is obtained by using linear matrix inequalities (LMIs) which simplifies the calculation of the contraction metric;
- Saturation constraint is incorporated into the LMI framework for contraction metric calculation;
- ...; and
- The effectiveness and the feasibility of the proposed method is validated through numerical simulations via **Lorenz system**.

2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorems.

2.1 Contraction Theory

Consider that we have the following deterministic nonlinear system:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t). \quad (1)$$

The contraction theory commences with the definition of *differential dynamics* of the system (1) using *differential displacement* $\delta\mathbf{x}$ (Kirk, 2004, Chap 4), which is defined as follows:

$$\frac{d}{dt}\delta\mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\delta\mathbf{x}. \quad (2)$$

The differential displacement $\delta\mathbf{x}$ is the infinitesimal displacement at *fixed time*.

The contraction property of the system (1) can be analyzed using the Lyapunov function $V := V(\mathbf{x}, t) = \frac{1}{2}\delta\mathbf{x}^\top \mathbf{M}\delta\mathbf{x}$, where $\mathbf{M} := \mathbf{M}(\mathbf{x}, \mathbf{x}_d, t) \in \mathbb{R}^{n \times n}$ the *contraction metric*. The contraction metric $\mathbf{M}$ is symmetric positive definite matrix and satisfies $\bar{m}\mathbf{I}_n \succeq \mathbf{M} \succeq m\mathbf{I}_n$ with $\bar{m}, m \in \mathbb{R}_{>0}$. For the system (1) to be contracting, the following condition must hold:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\frac{\partial \mathbf{f}}{\partial \mathbf{x}}) \leq -2\alpha\mathbf{M}, \quad (3)$$

which leads to $\frac{d}{dt}V \leq -\alpha V$ for some $\alpha \in \mathbb{R}_{>0}$. In other words, all solutions of the system (1) are contracting to a unique solution with an exponential rate α .

It is notable that the initial conditions are exponentially forgotten in the contracting system Lohmiller and Slotine (1998). Therefore, the contraction property is independent of the initial conditions.

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2.2 Partial Contraction Theory

On the other hand, the partial contraction theory investigates the contraction of the system using a virtual system. Following theorem details the partial contraction theory.

Theorem 1. (see Wang and Slotine (2004)). Consider a nonlinear system of the form $\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, \mathbf{x}, t)$ and the virtual system $\frac{d}{dt}\mathbf{q} = \mathbf{f}(\mathbf{q}, \mathbf{x}, t)$ are contracting with respect to $\mathbf{q}$. If a particular solution of virtual system verifies a smooth specific property, then all solutions of the original system verifies this property exponentially.

Example 1. Assume that the possible solutions of the virtual system defined in Theorem 1 are $\xi_1(t)$ and $\xi_2(t)$ which are the solutions of the nonlinear system $\mathbf{x}$.

2.3 Perturbed Contracting System

The following theorem presents the contraction property of the perturbed system by adding a bounded perturbation $\mathbf{d}(\mathbf{x}, t)$ to the system (1):

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{d}(\mathbf{x}, t). \quad (4)$$

Theorem 2. (Eq. 15 Lohmiller and Slotine (1998)). If the system (1) is contracting with the contraction metric $\mathbf{M}$ which satisfies the condition (3), the path integral $V_l(\mathbf{q}, \delta\mathbf{q}, t) = \int_{\xi_1}^{\xi_2} \|\Theta(\mathbf{q}, t)\delta\mathbf{q}\|$, where ξ_1 and ξ_2 are the solutions of (1) and (4), respectively, and $\mathbf{q}$ is the virtual [MS: ADD!!!]

$$\|\xi_1(t) - \xi_2(t)\| \leq \frac{V_l(0)}{\sqrt{m}} \exp(-\alpha t) + \frac{\bar{d}}{\alpha} \sqrt{\frac{m}{m}} (1 - \exp(-\alpha t)), \quad (5)$$

[MS: ADD!!!]

Proof. See Theorem 2.4 in Tsukamoto et al. (2021c).

[MS: Any more comments, if needed]

3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\begin{aligned} \frac{d}{dt}\mathbf{x} &= \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat}(\mathbf{u}) + \mathbf{d}(t) \\ &= \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t)\mathbf{u} + \boldsymbol{\phi}(\mathbf{u}) + \mathbf{d}(t), \end{aligned} \quad (6)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables, $\mathbf{u} \in \mathbb{R}^n$ denotes the control inputs, and $\mathbf{d}(t) \in \mathbb{R}^n$ denotes the external disturbances, and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ and $\mathbf{B} := \mathbf{B}(\mathbf{x}, t)$ denote system functions. The input saturation function is denoted by $\text{sat}(\cdot)$ and is defined as

$$\text{sat}(\mathbf{u}) = \begin{cases} \mathbf{u}, & \text{if } \|\mathbf{u}\| \leq \bar{u} \\ \bar{u} \frac{\mathbf{u}}{\|\mathbf{u}\|}, & \text{otherwise,} \end{cases} \quad (7)$$

where $\bar{u} \in \mathbb{R}_{>0}$ denotes the saturation threshold. The difference between the controller output $\mathbf{u}$ and the saturated control input $\text{sat}(\mathbf{u})$ is denoted by $\boldsymbol{\phi}(\mathbf{u}) = \text{sat}(\mathbf{u}) - \mathbf{u}$.

In the practical perspective, we consider the following assumption:

Assumption 1. The control input matrix $\mathbf{B}$ is sign-definite, so that the inverse of $\mathbf{B}$ exists.

Motivated by the use of the partial contraction theory in Tsukamoto et al. (2021a), [MS: and more...], we assume that the desired control input $\mathbf{u}_d \in \mathbb{R}^n$ which is designed to achieve the desired trajectory $\mathbf{x}_d \in \mathbb{R}^n$ under ideal conditions, i.e., $\mathbf{d} = 0$ and $\|\mathbf{u}_d\| \leq \bar{u}$, is given as follows:

$$\frac{d}{dt}\mathbf{x}_d = \mathbf{f}_d + \mathbf{B}_d\mathbf{u}_d, \quad (8)$$

where $\mathbf{f}_d := \mathbf{f}(\mathbf{x}_d, t)$ and $\mathbf{B}_d := \mathbf{B}(\mathbf{x}_d, t)$.

Finally, the control objective is to design a control input $\mathbf{u}$ such that the system trajectories of the system (4) and the desired system (8) converges, under the existences of the disturbances $\mathbf{d}$ and input saturation $\text{sat}(\cdot)$.

4. MAIN RESULTS

Following the control design in Tsukamoto et al. (2021a), the state-dependent coefficient (SDC) formulation Dani et al. (2015) is used to simplify the analysis to be understood as in LTV system theory. The SDC $\mathbb{A} := \mathbb{A}(\mathbf{x}, \mathbf{x}_d)$ is defined as follows:

$$\mathbb{A} := \int_0^1 \left[\frac{\partial \bar{\mathbf{f}}}{\partial \mathbf{x}} \right] (c\mathbf{x} + (1-c)\mathbf{x}_d) dc, \quad (9)$$

where $\bar{\mathbf{f}} := \bar{\mathbf{f}}(\mathbf{x}, \mathbf{x}_d) = \mathbf{f} - \mathbf{f}_d + \mathbf{B}\mathbf{u}_d - \mathbf{B}_d\mathbf{u}_d$ and c is a dummy variable for the integration. The existence... blah blah can be found in Dani et al. (2015), [MS: and more...].

Then, the error dynamics can be expressed as:

$$\begin{aligned} \frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) &= \mathbf{f} - \mathbf{f}_d + \mathbf{B}\mathbf{u} - \mathbf{B}_d\mathbf{u}_d + \mathbf{B}\boldsymbol{\phi}(\mathbf{u}) + \mathbf{d} \\ &= \underbrace{\mathbf{f} - \mathbf{f}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=\mathbb{A}(\mathbf{x}, \mathbf{x}_d)(\mathbf{x} - \mathbf{x}_d)} \\ &\quad + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\boldsymbol{\phi}(\mathbf{u}) + \mathbf{d} \\ &= \mathbb{A}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\boldsymbol{\phi}(\mathbf{u}) + \mathbf{d} \\ &= (\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M})(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}\boldsymbol{\phi}(\mathbf{u}) + \mathbf{d}, \end{aligned} \quad (10)$$

with the control input $\mathbf{u}$ defined as:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d) \quad (11)$$

where $\mathbf{R} = \mathbf{R}^\top \succ 0$ is a symmetric positive definite matrix.

In the following, we formulate the optimization problem to obtain the contraction metric $\mathbf{M}$ which satisfies the contraction condition (15) and does not violate the input saturation constraint (7).

4.1 Contraction Metric Calculation

Contraction Condition Satisfaction Invoking the partial contraction theory reviewed in Section 2.2, we can define the virtual system as follows:

$$\frac{d}{dt}\mathbf{q} = \frac{d}{dt}\mathbf{x}_d + (\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M})(\mathbf{q} - \mathbf{x}_d) + \mathbf{d}_\mu, \quad (12)$$

where $\mathbf{q} \in \mathbb{R}^n$ is the virtual state vector and $\mathbf{d}_\mu := \mathbf{d}_\mu(\mathbf{d}, \mu)$, $\forall \mu \in [0, 1]$, is defined as $\mathbf{0} = \mathbf{d}_\mu(\mathbf{d}, 0)$ and $\mathbf{d} = \mathbf{d}_\mu(\mathbf{d}, 1)$. The differential dynamics is then defined as follows:

$$\frac{d}{dt}\delta\mathbf{q} = (\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M})\delta\mathbf{q}. \quad (13)$$

[MS: CHECK FROM HERE] Then according to Theorem 1 and 2, system (12) and the system (8) will converge to the desired trajectory $\mathbf{x}_d$ with steady state error:

$$\mathbf{e}_{ss} = (\sup(\mathbf{B}\boldsymbol{\phi}(\mathbf{u})) + \bar{d}) \frac{1}{\alpha} \sqrt{\frac{m}{m}}, \quad (14)$$

with the following conditions:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\mathbb{A}) + 2\mathbf{M}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \leq -2\alpha\mathbf{M}. \quad (15)$$

By multiplying $\mathbf{W} := \mathbf{M}^{-1}$ both sides of (15), we have the following LMI:

$$-\frac{d}{dt}\mathbf{W} + 2\text{sym}(\mathbb{A}\mathbf{W}) - 2\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \leq -2\alpha\mathbf{W}. \quad (16)$$

In [MS: []], Tsukamoto et al. proposed a method to formulate the LMI problem for the contraction metric as follows:

$$\begin{aligned} \min_{\mathbf{W}, X, \mu} \quad & X + \lambda\mu \\ \text{s.t.} \quad & (15) \end{aligned} \quad (17)$$

However, even with the proposed method, the input saturation can still occur by excessive contraction metric leading to potential instability. Therefore, we need to incorporate input constraints into the LMI formulation.

Input Saturation Constraint To address the input saturation function (7), the saturation constraint $\mathbf{u}^\top \mathbf{u} - \bar{u}^2 \leq 0$ is defined as:

$$\begin{bmatrix} \bar{u}^2 & (\mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d))^\top \\ \star & \mathbf{I} \end{bmatrix} \succeq 0. \quad (18)$$

By multiplying both sides by $[\text{diag}(\mathbf{I}, \mathbf{W}(\mathbf{R}^{-1}\mathbf{B}^\top)^{-1})]$, we have LMIs as follows:

$$\begin{bmatrix} \bar{u}^2 & (\mathbf{W}(\mathbf{R}^{-1}\mathbf{B}^\top)^{-1}\mathbf{u}_d - (\mathbf{x} - \mathbf{x}_d))^\top \\ \star & \mathbb{X} \end{bmatrix} \succeq 0, \quad (19)$$

where $\mathbb{X} := \mathbf{W}(\mathbf{R}^{-1}\mathbf{B}^\top)^{-1}(\mathbf{R}^{-1}\mathbf{B}^\top)^{-1}\mathbf{W}$ which satisfies

$$\begin{bmatrix} \mathbb{X} & \mathbf{W}((\mathbf{R}^{-1}\mathbf{B}^\top)^{-1})^\top \\ \star & \mathbf{I} \end{bmatrix} \succeq 0. \quad (20)$$

Finally, the contraction metric can be obtained by solving the following LMI problem:

$$\begin{aligned} \min_{\mathbf{W}, X, \mu} \quad & X + \text{trace}(\mathbb{X}) + \lambda\mu \\ \text{s.t.} \quad & (15), (19) \text{ and } (20) \end{aligned} \quad (21)$$

4.2 Numerical Methods For LMI

- Discretize $\frac{d}{dt}\mathbf{M}$
- LMI solver
- Tuning methods

5. NUMERICAL VALIDATION

5.1 Numerical Validation Set Up

We employed the Lorenz system [MS: REF] to validate the proposed method which is represented as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \xi} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}}_{=: f(\xi)} + \underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: \mathbf{u}} + \underbrace{\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}}_{=: \mathbf{d}}, \quad (22)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters.

5.2 Numerical Validation Results

6. CONCLUSION

In this study, ...

ACKNOWLEDGEMENTS

Place acknowledgments here.

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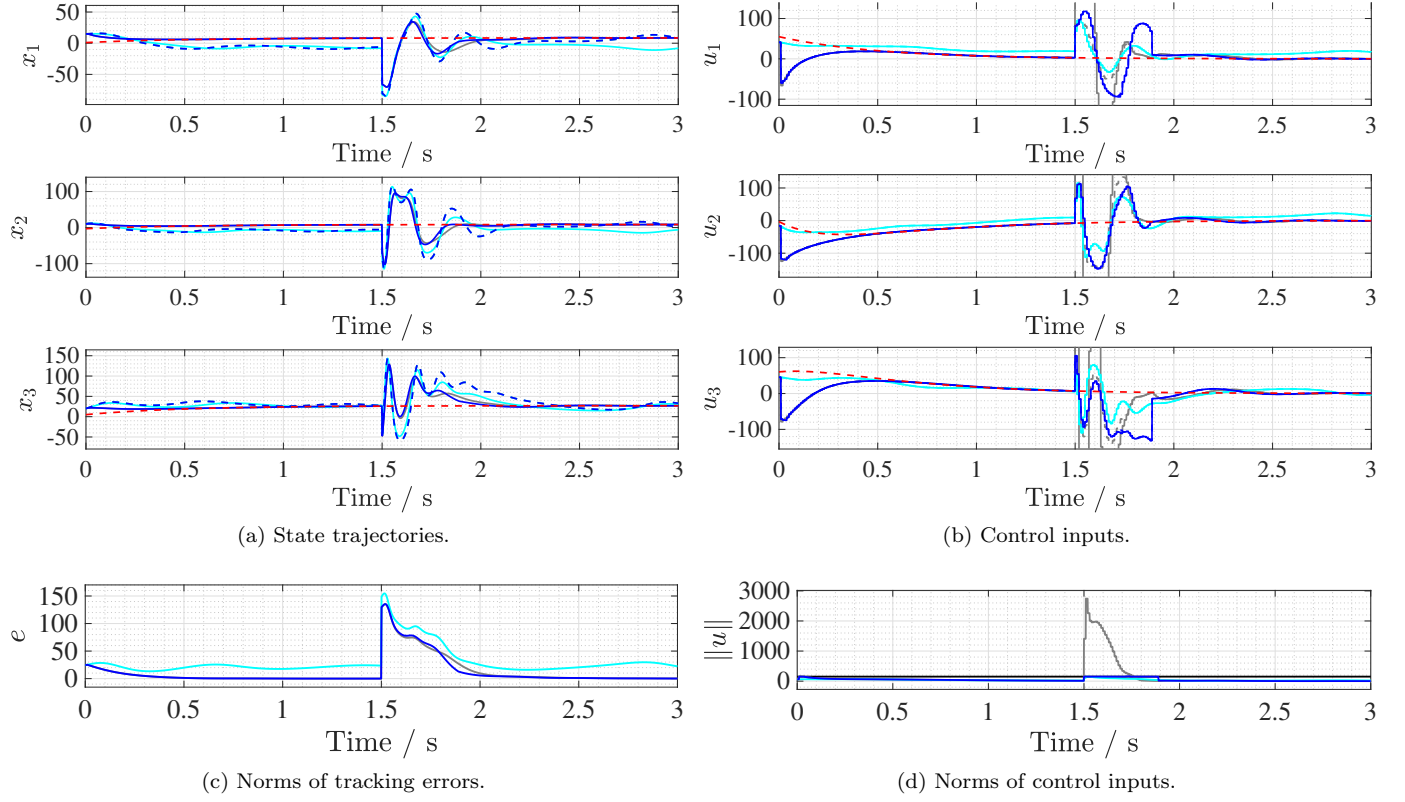


Fig. 1. Numerical validation results of (C_1) [—], (C_2) [—], (C_3) [—], (C_4) [—], and desired trajectory x_d [---].

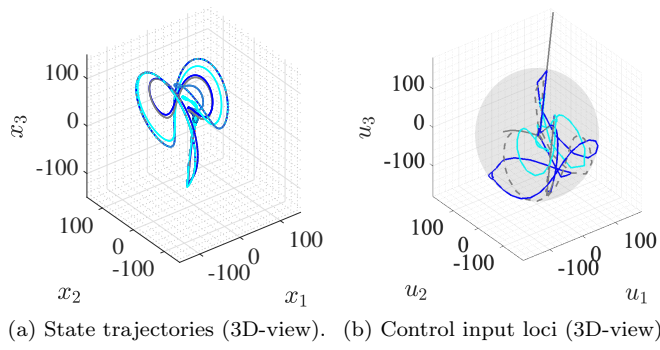


Fig. 2. Numerical validation results of (C_1) [—], (C_2) [—], (C_3) [—], (C_4) [—], and desired trajectory x_d [---].